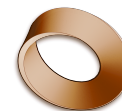


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.8 – Radio & Microwaves

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain how radio waves are produced and used for communication.
- b) I can distinguish between the uses and power levels of microwaves in ovens versus mobile devices.
- c) I can evaluate scientific evidence regarding the health risks of non-ionising radiation from everyday technology.

Theory

Radio waves are a type of electromagnetic wave with the longest wavelengths in the spectrum. They are typically produced when **electrons** oscillate back and forth in a metal wire, known as an antenna. This movement creates a changing electric and magnetic field that propagates through space. We use radio waves for broadcasting because they can travel long distances and pass through buildings. By encoding information onto these waves, we can transmit sound for radio stations, video for television, and data for **walkie-talkies**.

Microwaves sit just above infrared waves in the electromagnetic spectrum. They have shorter wavelengths and higher frequencies than radio waves. Microwaves are essential for modern technology; they are used to carry data for **Wi-Fi** networks and mobile phone signals. Because microwaves can penetrate clouds and smoke, they are also used in satellite communications. However, their most well-known domestic use is in microwave ovens, where they cause water molecules in food to vibrate rapidly, generating heat through a process called dielectric heating.

It is crucial to understand the difference in power between these devices. A standard mobile phone or Wi-Fi router emits microwaves at very low power levels, measured in watts (often less than 1 watt). In contrast, a microwave oven operates at a power of roughly 800 to 1200 watts. This means the radiation inside an oven is thousands of times more powerful than that from your phone. Microwave ovens are designed with metal mesh screens in the door to contain these high-power waves, preventing them from escaping and cooking you accidentally.

The question of whether low-power microwaves from phones and Wi-Fi pose a health risk is a topic of ongoing scientific debate. This falls under **non-ionising radiation**, which means the waves do not have enough energy to remove electrons from atoms or damage DNA directly, unlike X-rays. Some people worry about long-term exposure, while others argue that since these devices are low power, they are safe. Scientists look for **evidence** by studying large groups of people over many years to see if there is a link between device use and health issues like cancer. Currently, major health organisations state that there is no conclusive proof that low-level microwave exposure causes harm, but research continues to monitor any potential effects.

Word bank: walkie-talkies · electrons · Wi-Fi · evidence · non-ionising

Activity

Investigating Signal Penetration

Work in pairs. You will need two mobile phones and a thick book or a metal box (like a biscuit tin with the lid on).

- Call Phone A from Phone B. Note that the call is clear.
- Place Phone A inside the metal box and close the lid tightly. Try calling it again. What happens to the signal? Explain why this happens using the concept of **electromagnetic**

shielding.

- Now, place Phone A under a thick pile of books or a wet towel. Call it again. Does the signal weaken? Why might water or dense materials affect radio waves differently than metal?
- Record your observations in the table below.

Material	Signal Strength (Strong/Medium/Weak/None)	Explanation (Why did this happen?)
Air (Control)		
Metal Box		
Books/Water		

Questions

1. Explain in your own words how a radio station uses electrons to send music to your car radio.

2. Why is a microwave oven thousands of times more dangerous to human tissue than a mobile phone if you were exposed to it directly?

3. Define **non-ionising radiation** and explain why this term is important when discussing the safety of Wi-Fi routers.

4. Some people believe that putting your phone near your bed while sleeping is unsafe because of microwave emissions. Using the concept of **evidence**, describe how you would investigate this claim scientifically rather than just relying on opinions.

Extension

Research the "Precautionary Principle." If scientists are not 100% sure that low-level microwave exposure is harmless, should we ban these technologies? Write a short paragraph arguing for or against banning Wi-Fi in schools based on scientific evidence and practical needs.
